

# Applied Robotics Laboratory: Conveyor Tracking

## Hardware Overview and Documentation

### 1 Mechanical Setup

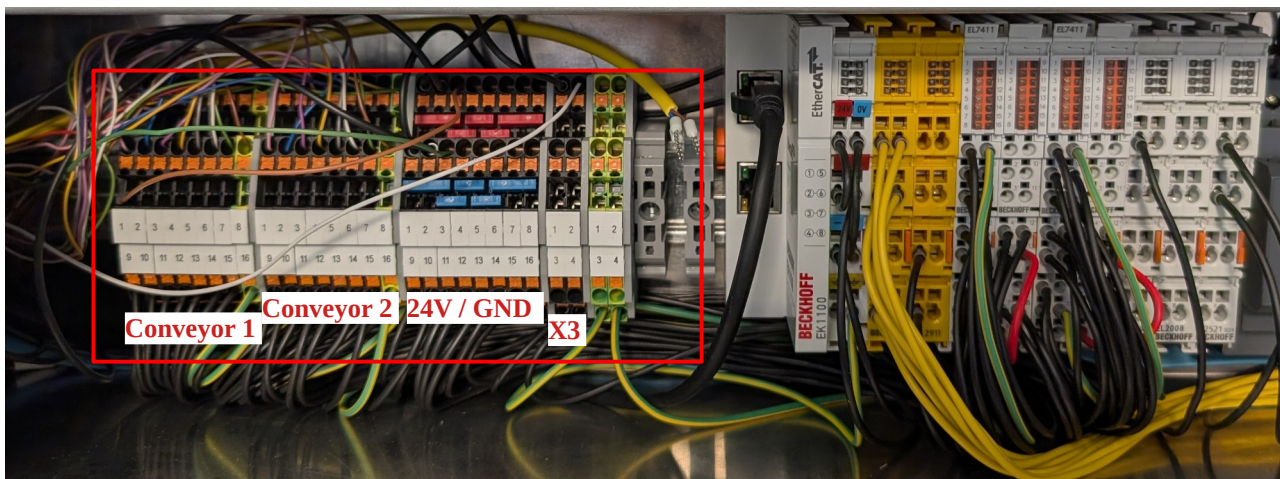
CAD is handed in separately and includes all 3D-Printed parts (STL/STEP).

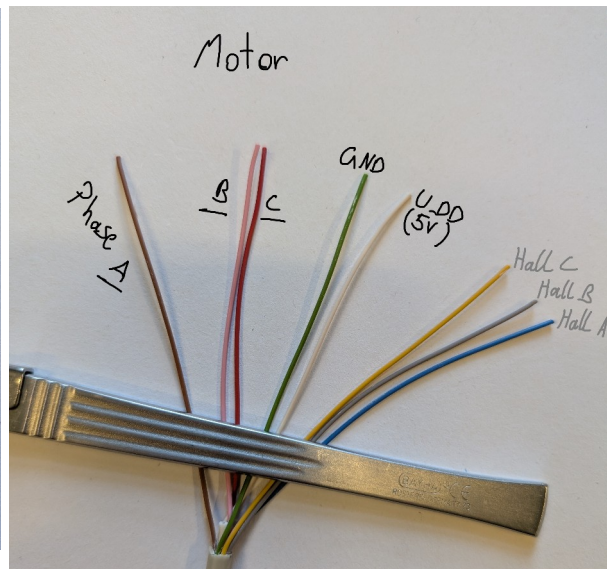
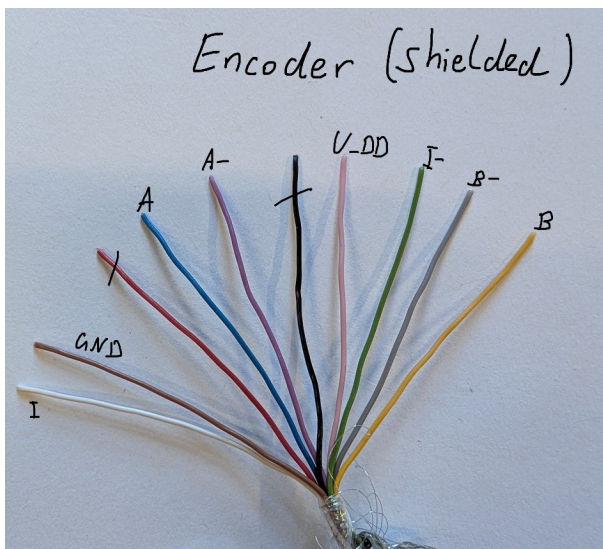
### 2 PLC Setup

#### 2.1 Wiring

| Component | Usage                                | Notes                                                    |
|-----------|--------------------------------------|----------------------------------------------------------|
| EK1100    | EtherCAT                             |                                                          |
| EL2911    | Safety                               |                                                          |
| EL7411    | BLDC Drive Conveyor                  |                                                          |
| EL7411    | BLDC Drive Conveyor                  |                                                          |
| EL1014    | Digital Input: Hall Sensor           |                                                          |
| EL2008    | Digital Outputs: R/G LED             | On old setup: ENC simulation via pulsing digital outputs |
| EL2521    | Encoder Simulation Output to DSQC377 | Axis Slave of EL7411                                     |

All connections are broken out:





Encoder Channel A, B and C (or I) must all be connected to EL7411.

The Hall Sensor of the motor is not required but still has been connected.

Only minor changes have been made to the electrical system after the first EPLAN was made. It might be useful to update the EPLAN to represent the current state.

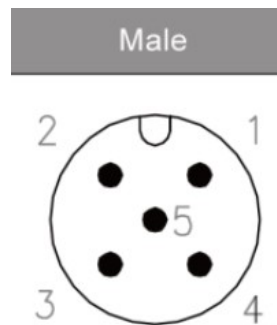
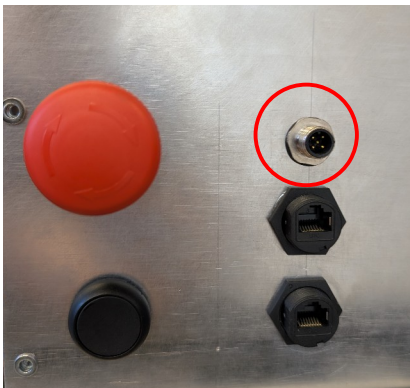
The Terminal Assignment can be seen on the next page.

|  |                 |                |                   |  |  |                 |
|--|-----------------|----------------|-------------------|--|--|-----------------|
|  |                 |                |                   |  |  |                 |
|  |                 |                | <b>Conveyor 1</b> |  |  |                 |
|  | <b>Motor</b>    | UDD            | 1                 |  |  | 7               |
|  |                 | Phase A        | 2                 |  |  | 4°              |
|  |                 | Phase B        | 3                 |  |  | 12°             |
|  |                 | Phase C        | 4                 |  |  | 5°              |
|  |                 | Hallsensor A   | 5                 |  |  | 5               |
|  |                 | Hallsensor B   | 6                 |  |  | 6               |
|  |                 | Hallsensor C   | 7                 |  |  | 13              |
|  |                 | GND            | 8                 |  |  | 15              |
|  | <b>Encoder</b>  | UDD            | 9                 |  |  | 1               |
|  |                 | Kanal A+       | 10                |  |  | 2               |
|  |                 | Kanal A-       | 11                |  |  | 10              |
|  |                 | Kanal B+       | 12                |  |  | 3               |
|  |                 | Kanal B-       | 13                |  |  | 11              |
|  |                 | Kanal C+       | 14                |  |  | 4               |
|  |                 | Kanal C-       | 15                |  |  | 12              |
|  |                 | GND            | 16                |  |  | 9               |
|  | <b>EL7411.1</b> | 13°            |                   |  |  | 8°              |
|  |                 |                |                   |  |  | <b>EL7411.1</b> |
|  |                 |                | <b>Conveyor 2</b> |  |  |                 |
|  | <b>Motor</b>    | UDD            | 1                 |  |  | 7               |
|  |                 | Phase A        | 2                 |  |  | 4°              |
|  |                 | Phase B        | 3                 |  |  | 12°             |
|  |                 | Phase C        | 4                 |  |  | 5°              |
|  |                 | Hallsensor A   | 5                 |  |  | 5               |
|  |                 | Hallsensor B   | 6                 |  |  | 6               |
|  |                 | Hallsensor C   | 7                 |  |  | 13              |
|  |                 | GND            | 8                 |  |  | 15              |
|  | <b>Encoder</b>  | UDD            | 9                 |  |  | 1               |
|  |                 | Kanal A+       | 10                |  |  | 2               |
|  |                 | Kanal A-       | 11                |  |  | 10              |
|  |                 | Kanal B+       | 12                |  |  | 3               |
|  |                 | Kanal B-       | 13                |  |  | 11              |
|  |                 | Kanal C+       | 14                |  |  | 4               |
|  |                 | Kanal C-       | 15                |  |  | 12              |
|  |                 | GND            | 16                |  |  | 9               |
|  | <b>EL7411.2</b> | 13°            |                   |  |  | 8°              |
|  |                 |                |                   |  |  | <b>EL7411.2</b> |
|  |                 |                | <b>X10</b>        |  |  |                 |
|  | <b>24V</b>      | SPSV+          | 1                 |  |  |                 |
|  |                 | IND Sensor 24V | 2                 |  |  | IPC 24V         |
|  |                 |                | 3                 |  |  | A4-BC1 24V      |
|  |                 |                | 4                 |  |  | EL7411.1 7°     |
|  |                 |                | 5                 |  |  | EL7411.2 7°     |
|  |                 |                | 6                 |  |  | EL2911 4°       |
|  |                 |                | 7                 |  |  |                 |
|  |                 |                | 8                 |  |  |                 |
|  | <b>0V</b>       | SPSV-          | 9                 |  |  | LS GND          |
|  |                 |                | 10                |  |  | IPC GND         |
|  |                 |                | 11                |  |  | A4-BC1 GND      |
|  |                 |                | 12                |  |  | EL 7411.1 15°   |
|  |                 |                | 13                |  |  | EL 7411.2 15°   |
|  |                 |                | 14                |  |  | EL 2911 3°      |
|  |                 |                | 15                |  |  |                 |
|  |                 | IND Sensor GND | 16                |  |  |                 |
|  |                 |                |                   |  |  |                 |
|  |                 |                | <b>X20</b>        |  |  |                 |
|  |                 | IND Sensor NO  | 1                 |  |  | EL 1014 1 IND   |
|  |                 |                | 2                 |  |  | LS              |
|  |                 |                | 3                 |  |  |                 |
|  |                 |                | 4                 |  |  |                 |
|  |                 |                |                   |  |  |                 |
|  |                 |                | <b>X30</b>        |  |  |                 |
|  | <b>PE</b>       |                | 1                 |  |  | A4-BC1 8        |
|  |                 |                | 2                 |  |  |                 |
|  |                 |                | 3                 |  |  |                 |
|  |                 |                | 4                 |  |  |                 |

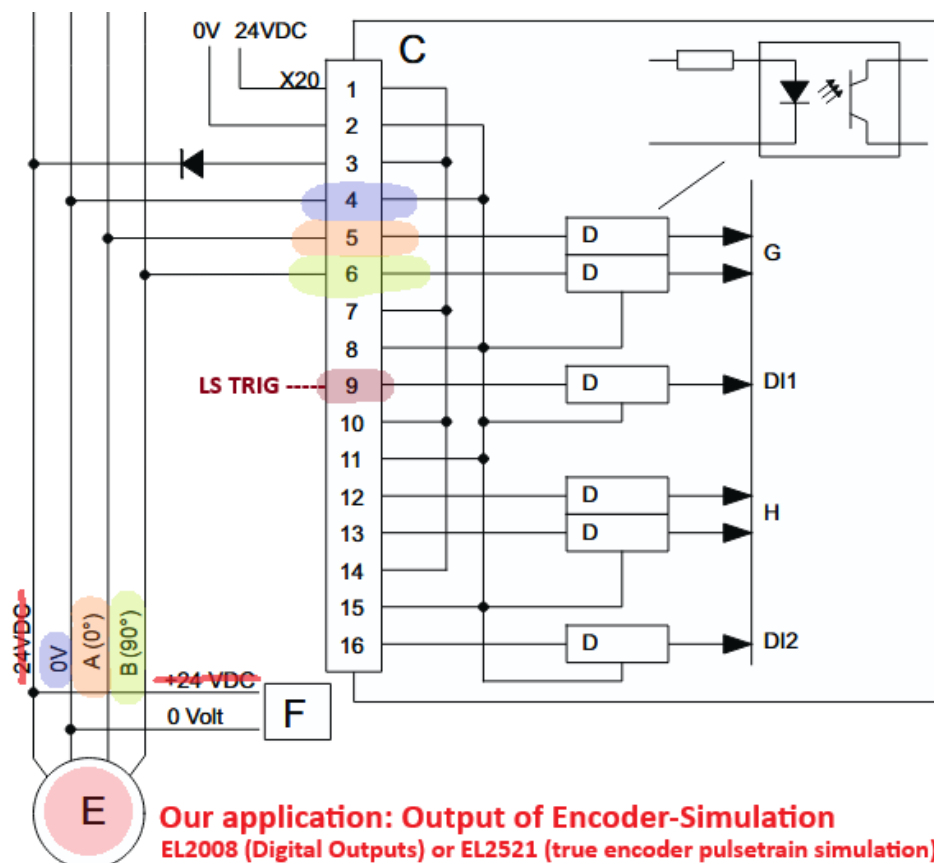
## 2.2 Electrical Signals PLC → Robot

### *M12 5-Pin Connector for ENC and TRIG signal → Conveyor Tracking*

| Pin | Usage    |
|-----|----------|
| 1   | EncA     |
| 2   | EncB     |
| 3   | GND 0V   |
| 4   | None     |
| 5   | +LS TRIG |



See ABB Document 3HAC050991-001 (Application Manual Conveyor Tracking) for detailed description on the DSQC377 card. Here is an overview of the electrical connections.



## 2.3 Network Connection PLC → Robot

PROFINET – use PN-Backups of Robot controllers

SPS\_67 only connects to robot ...67 Wi-Fi, and SPS\_68 only to ...68.

Both PLC have the static IP address 192.168.125.5 → each in the respective network. If you want to connect to the PLC via remote desktop, set your own adapter IP settings to automatic and the Wi-Fi router will give you the right address via DHCP. If you use manual addresses, use 192.168.125.XXX but not .5!

Adapter settings are automatic for PROFINET and EtherCAT – no static addresses!

## 2.4 Encoder Signal generation from PLC to Robot

On SPS\_67, the EL2521 is used (**not yet finally working in the program**) to generate a 24V pulse (encoder) signal from the speed of the EL7411 Camera\_Axis.

On SPS\_68, the EL2008 Digital Output card is used to generate a 24V pulse signal with the delegated task and Main\_Encoder program.

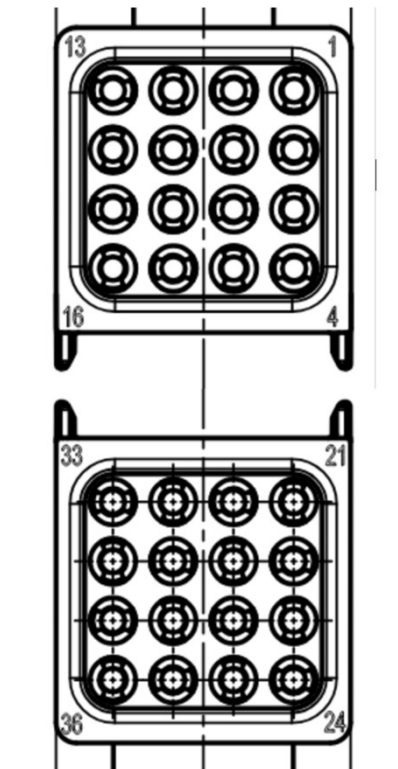
Both have the same purpose: Generate a 24V encoder signal for the DSQC377 (ABB Conveyor Tracking) to process. **The DSQC377 needs a minimum level of 18V but the encoder in the conveyors is TTL 5V.** The solution with EL2521 is the correct, recommended way to solve this issue. Using digital outputs is only a temporary workaround.

### 3 HARTING

On the old setup, the two modules are separate and a connection is established using the HARTING industrial connector. The “Han Gelenkrahmen Plus” frame holds 4 “Han Domino” cubes. Power transmission for the camera, power and signal for the light barrier (LS) sensor and all signals for the camera conveyor are routed via two “Han Domino HD” cubes – each with 16 pins, so a total of 32 pins is available on the HARTING connector. Conveniently, the 16-pin cubes have labels 1-16 and 21-36 for each pin, as visible on the image. The pin occupation is as follows:

***HARTING Pin Occupation***

| Pin   | Usage                                        |
|-------|----------------------------------------------|
| 1-10  | Encoder (passthrough, wires see chapter 2.1) |
| 11    | +24V (for LS and Camera)                     |
| 12    | +Out LS (Trig)                               |
| 13    | GND 0V                                       |
| 14    | -                                            |
| 15    | -                                            |
| 16    | -                                            |
| 21-28 | Motor (passthrough, wires see chapter 2.1)   |
| 29-36 | -                                            |

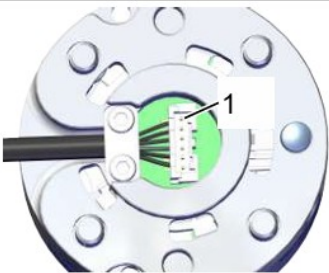


Ethernet for the COGNEX camera is passed through the HARTING connector as well.

## 4 SCHMALZ

Both Vacuum generators were delivered with cables attached. SCHMALZ uses JST-PA style (RS: 476-6805) connectors. The original cables were removed and a short cable for the quick-change electrical passthrough was soldered to a JST-PA connector. This was done so that the original SCHMALZ cables don't need to be cut. The wiring can be seen here:

**PIN-Belegung Flanschstecker 6-polig**

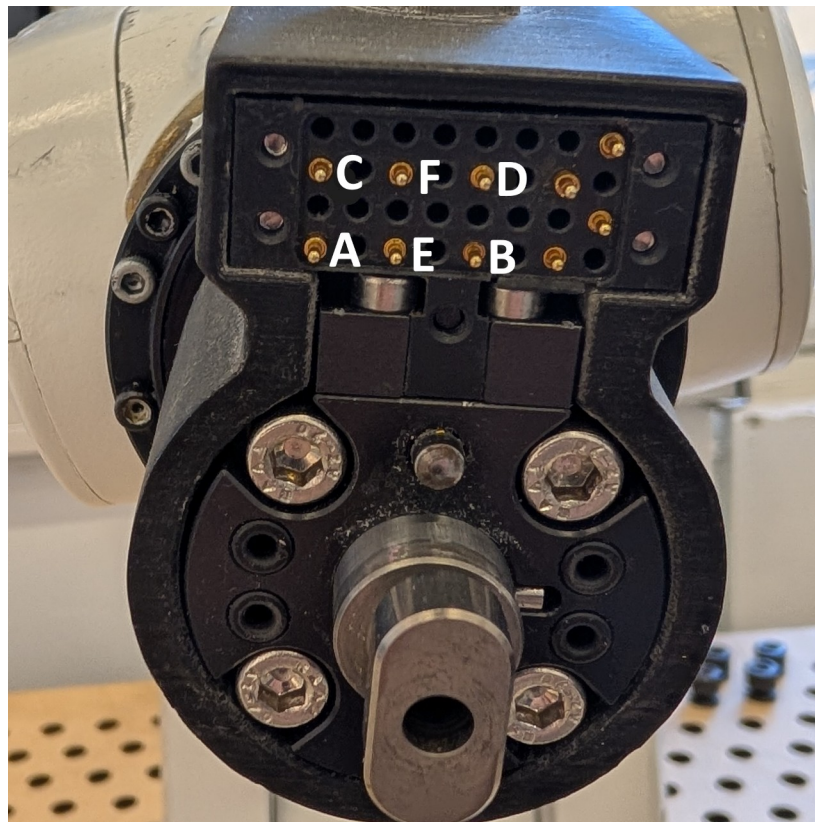
| Flanschstecker                                                                    | PIN | Symbol | Funktion                                           |
|-----------------------------------------------------------------------------------|-----|--------|----------------------------------------------------|
|  | 1   | IN2    | Signaleingang „Ablegen“                            |
|                                                                                   | 2   | IN1    | Signaleingang „Saugen“                             |
|                                                                                   | 3   | OUT3   | Optionaler Signalausgang z.B. Freedrive aktivieren |
|                                                                                   | 4   | OUT2   | Signalausgang "Part Present" / IO-Link             |
|                                                                                   | 5   | GND    | Masse                                              |
|                                                                                   | 6   | U      | Versorgungsspannung 24 V                           |

The connections must match the pins of the quickchange system on the robot. See the yellow folder in the lab office for a detailed layout of the quickchanger.

**SCHMALZ Quickchange Pins**

| Quickchange Pin | Function    |
|-----------------|-------------|
| A               | +24V        |
| B               | DO1 (DO07)  |
| C               | GND 0V      |
| D               | DO2 (DO08)  |
| E               | DI1 (DI07?) |
| F               | DI2 ?       |

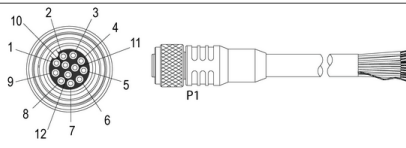
**Watch out! The pins on the receiving end of the quick changer are mirrored!**



The vacuum grippers are type SFB1 10 SI-55 M5-AG

## 5 COGNEX

The camera tower is detachable from the base. The camera is connected via a CCB-M12X12FS-05 cable to the breakout wago clips inside the camera tower.



| Pin# | Signal Names                                                               | Wire Color |
|------|----------------------------------------------------------------------------|------------|
| 1    | Out 2/In 2                                                                 | White      |
| 2    | TxD                                                                        | Green      |
| 3    | RxD                                                                        | Pink       |
| 4    | Out 3/In 3                                                                 | Yellow     |
| 5    | In 1                                                                       | Grey       |
| 6    | Common In <span style="color: red;">pull to ground for PNP trigger!</span> | Black      |
| 7    | +24 VDC                                                                    | Brown      |
| 8    | GND                                                                        | Blue       |
| 9    | Common Out                                                                 | Purple     |
| 10   | In 0 <span style="color: red;">+LS TRIG (PNP)</span>                       | Red        |
| 11   | Out 0                                                                      | Grey/Pink  |
| 12   | Out 1                                                                      | Red/Blue   |

Important: Look closely at the label of the COGNEX cable, there are different models with different wire colors!!!

At the bottom of the camera tower, as well as at the base, an A-coded M12 8 pin connector is used to pass through the electrical connections for the camera:

### *COGNEX Cable → Camera Tower*

| Function | Cognex Cable | M12x8 Pin Cable | M12x8 Pin |
|----------|--------------|-----------------|-----------|
| +24V     | Brown        | Brown           | 2         |
| GND 0V   | Blue + Black | Blue            | 7         |
| +LS TRIG | Red          | Yellow          | 4         |

M12 8 pin connector matches to the camera tower base which has labeled wires and connects via Wago clips to the rest of the system from above.

**Ethernet for the camera is passed through the whole system down into the Wi-Fi router of the robot!** The robot uses its native network protocol for communication with the camera.

## 6 Bill of Materials

The following table gives an overview of important, specifically sourced components used. It does not claim to be complete and many of the components used and not listed are normally in stock at the MCI robotics lab.

| Name                                              | Usage in Component Group        | Count per Setup | RS Number or Manufacturer Part Number |
|---------------------------------------------------|---------------------------------|-----------------|---------------------------------------|
| Festo Light Barrier                               | Camera Tower Base               | 1               | SOOD-TB-R-PN                          |
| Eaton LED R/G                                     | Part-Conveyor Cover             | 1               | 180799 M22-FLED-RG                    |
| Eaton Push Button                                 | Main                            | 1               | M22S-D-S 216591                       |
| Eaton E-Stop Twist Release                        | Main                            | 1               | 263467 M22-PVT                        |
| MeanWell 24V 6.25A PSU                            | Main                            | 1               | HDR-150-24                            |
| RS PRO RJ45-Coupler Panelmount Straight           | Main                            | 2               | 111-6746                              |
| RS PRO RJ45-Coupler Panelmount 90°                | Camera Tower Base, Camera Tower | 2               | 111-6747                              |
| RS PRO M12 5-pin Male Connector Panelmount        | Main                            | 1               | 877-1154                              |
| RS PRO M12 8-Pin Male Connector Panelmount        | Camera Tower Base               | 1               | 208-0595                              |
| RS PRO M12 8-Pin Female Connector Panelmount      | Camera Tower                    | 1               | 208-0582                              |
| RS PRO M12 8-Pin Male Connector for Cable         | Connection Cable Camera Tower   | 1               | 208-0570                              |
| RS PRO M12 8-Pin Female Connector with open Cable | Connection Cable Camera Tower   | 1               | 206-7628                              |
| Assemtech Inductive Sensor                        | Main                            | 1               | 866-2628                              |
| Vacuum Gripper                                    | SCHMALZ                         | 2               | SFB1 10 SI-55 M5-AG                   |